

UNIT-5 BaseBand Shaping for Data Transmission

11

Discrete PAM signals:

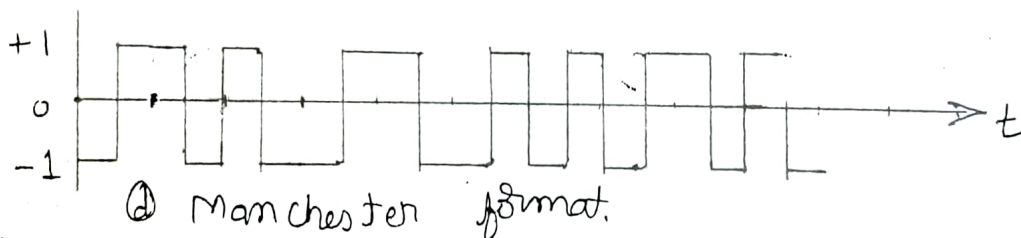
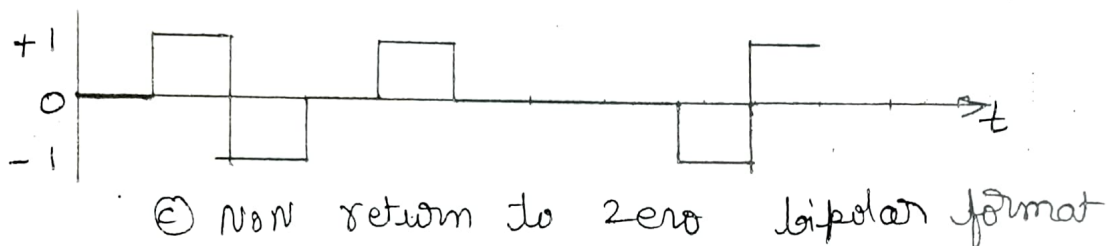
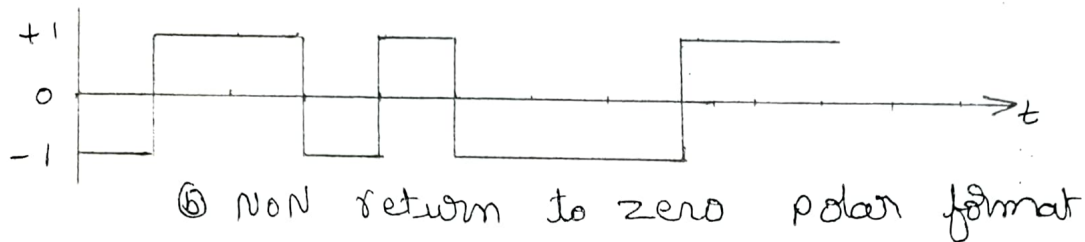
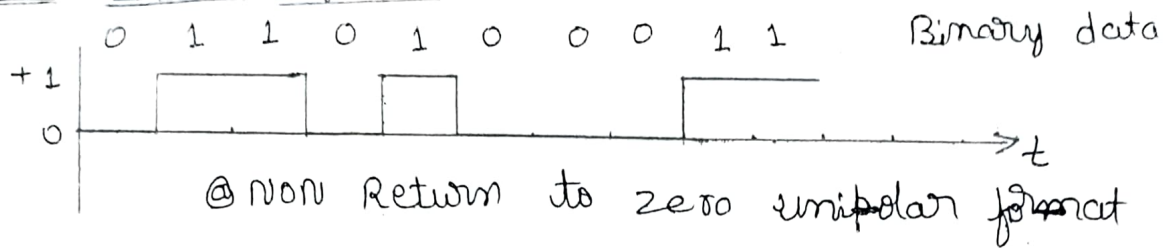


fig: Binary data formats

The use of an appropriate waveform for baseband representation of digital data is basic to its transmission from a source to a destination. fig shows four different formats for the representation of the binary data sequence 0110100011.

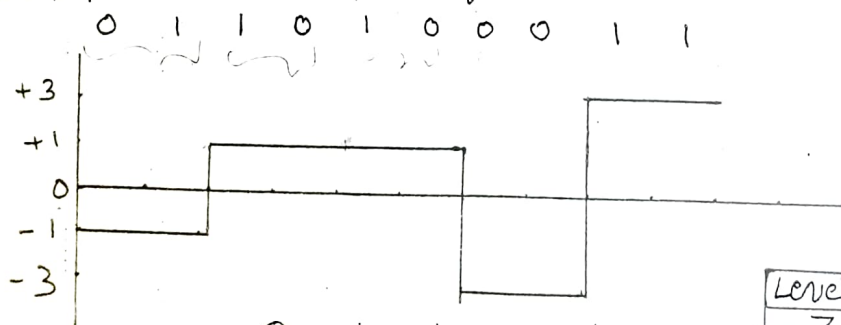
In the unipolar format, symbol 1 is represented by transmitting a pulse, whereas symbol 0 is represented by switching off the pulse. When the pulse occupies the full duration of a symbol the unipolar format is said to be of the non-return to zero (NRZ) type. When it occupies a fraction of the symbol duration, it is said to be of return to zero (RZ) type.

In the polar format, a positive pulse is transmitted for symbol 1 and a negative pulse for symbol 0. It can be of NRZ or RZ type. The NRZ version of the polar format is shown in fig (b). Unlike the unipolar waveform, a polar waveform has no dc component.

In the bipolar format, positive and negative pulses are used alternately for the transmission of 1's and no pulse for the transmission of 0's. It can be of NRZ or RZ type. The NRZ version of the bipolar format is shown in fig (c). The puls alternating property of the bipolar format provides a capability for in-service performance monitoring in the sense that any isolated error, whether it causes the deletion or creation of a pulse, will violate this property.

In the Manchester format (biphase base band signalling), symbol 1 is represented by transmitting a positive pulse for one-half of the symbol duration, followed by a negative pulse for the remaining half of the symbol duration; for symbol 0 the two pulses are transmitted in reverse order. It is in fig (d).

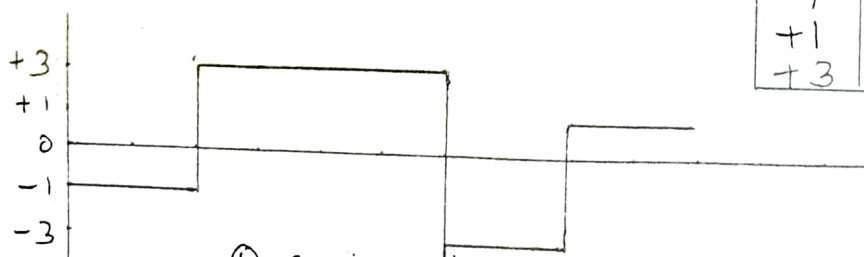
The NRZ versions of unipolar, polar & bipolar formats make efficient use of Bandwidth, but they do not offer synchronization capability. Manchester format has a built in synchronization capability because there is a predictable transition during each bit interval. But this capability is attained at the expense of a bandwidth requirement twice that of the NRZ unipolar, polar and bipolar formats.



(a) Natural encoded

Level	Natural code	Gray code
-3	00	00
-1	01	01
+1	10	10
+3	11	10

Table



(b) Gray - encoded

fig: Polar quaternary (M-ary format)

The utilisation of Bandwidth can be made more efficient by adopting an M -ary format for the representation of the input binary data. The polar quaternary format of the NRZ type is shown in fig. A quaternary code has four distinct symbols referred to as dibits (Pair of bits). Each dibit is assigned a level in accordance with the natural code or gray code as in table. The input binary sequence 0110100011 is viewed as a new sequence of dibits, {01, 10, 10, 00, 11} and each dibit in the sequence is represented by a level of its own. In Gray code the adjacent bits are arranged in such a way that they differ by only one bit.

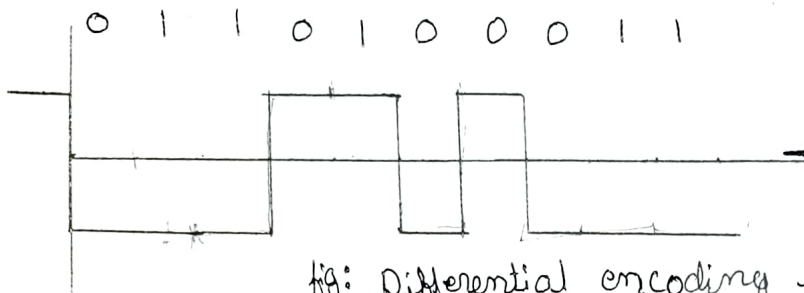


fig: Differential encoding format

Differential encoding starts with an arbitrary initial bit. In fig a positive level is shown for the initial bit. The information in the input binary sequence itself is encoded in terms of signal transition. For e.g. a signal transition is used to designate symbol 0, while no transition is used to designate symbol 1. The polarity of a differentially encoded waveform may be inverted without affecting its interpretation. The original binary information may be recovered by sampling the received wave and comparing the polarity of adjacent samples to establish whether or not a signal transition has occurred.

Power Spectra of Discrete PAM Signals

NRZ format	co-efficient A_k	Basic Pulse $v(t)$
Unipolar	$A_k = \begin{cases} a, \text{ Symbol 1} \\ 0, \text{ Symbol 0} \end{cases}$	$v(t)$ consists of rectangular pulse of unit amplitude and duration T_b
Polar	$A_k = \begin{cases} a, \text{ Symbol 1} \\ -a, \text{ Symbol 0} \end{cases}$	
Bipolar	$A_k = \begin{cases} a, -a, \text{ alternating 1s} \\ 0, \text{ Symbol 0} \end{cases}$	
Manchester -er	$A_k = \begin{cases} a, \text{ Symbol 1} \\ -a, \text{ Symbol 0} \end{cases}$	$v(t)$ consists of doublet pulse of heights ± 1 and total duration T_b
Polar quaternary (natural coding)	$A_k = \begin{cases} 3a, \text{ dibit 11} \\ a, \text{ dibit 10} \\ -a, \text{ dibit 01} \\ -3a, \text{ " 00} \end{cases}$	$v(t)$ consists of rectangular pulse of unit amplitude and duration $2T_b$

The signalling format is the implementation of a discrete amplitude-modulated pulse train. The different realisations of a random process $X(t)$ defined by

$$X(t) = \sum_{k=-\infty}^{\infty} A_k v(t-kT) \quad \text{--- (1)}$$

where the co-efficient A_k is a discrete random variable, $v(t)$ is a basic pulse shape and T is the symbol duration.

The power spectral density of the discrete PAM signal $X(t)$ is given by $S_X(f) = \frac{1}{T} |V(f)|^2 \sum_{n=-\infty}^{\infty} R_A(n) \exp(-j2\pi f n T)$ --- (2)

where $V(f)$ is the Fourier transform of the basic pulse $v(t)$. $R_A(n)$ is the auto-correlation function given as $R_A(n) = E[A_k A_{k-n}]$ where E is the expectation operation. $R_A(n)$ and $V(f)$ function values depend on the type of discrete PAM signal. The evaluations for other other formats are given below.

1 NRZ unipolar Format:

Suppose the 0's and 1's of a random binary sequence occur with equal probability. Then for a unipolar format of the NRZ type, we have

$$P(A_k=0) = P(A_k=a) = 1/2$$

Hence, for $n=0$, autocorrelation function is given as

$$E[A_k^2] = (0)^2 P(A_k=0) + (a)^2 P(A_k=a) = a^2/2$$

consider next the product $A_k A_{k-n}$ for $n \neq 0$. This product has four possible values, 0, 0, 0 and a^2 i.e. $[0, 0, 0, a^2]$. Assuming that the successive symbols in the binary sequence are statistically independent, these four values occur with a probability of $1/4$ each. Hence, for $n \neq 0$, we may write

$$E[A_k A_{k-n}] = 3(0) \frac{1}{4} + (a^2) \left(\frac{1}{4}\right) = \frac{a^2}{4} \quad n \neq 0$$

we may thus express the auto correlation function $R_A(n)$ as

$$R_A(n) = \begin{cases} a^2/2 & n=0 \\ a^2/4 & n \neq 0 \end{cases} \quad \text{--- (3)}$$

For the basic pulse $v(t)$, we have a rectangular pulse of unit amplitude and duration T_b . Hence, the Fourier transform of $v(t)$ equals

$$V(f) = T_b \text{sinc}(f T_b) \quad \text{--- (3)}$$

where the sinc function is defined by

$$\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$$

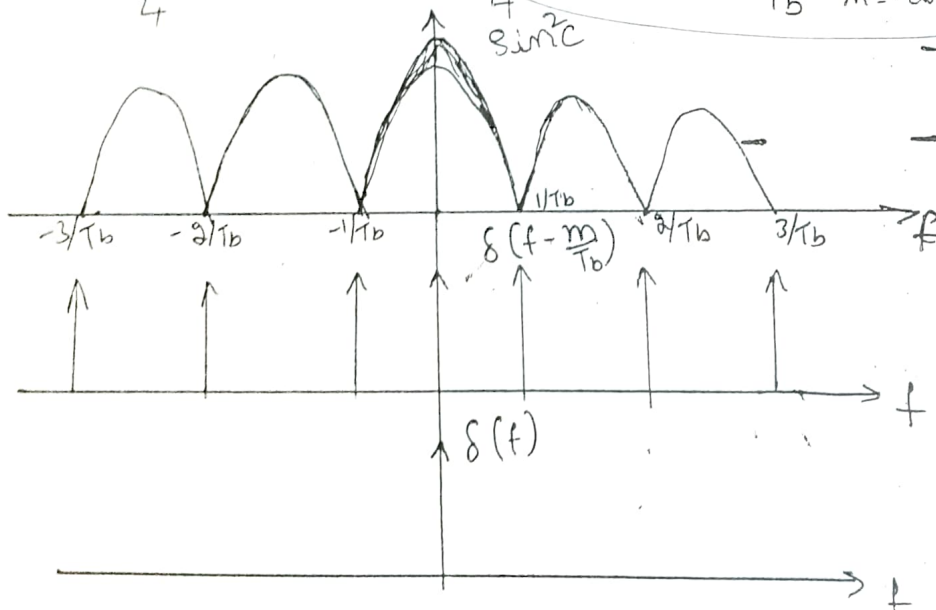
Hence using Eq (2) and (3) in Eq (1), with $T = T_b$, yields the following result for the power spectral density of the NRZ unipolar format:

$$S_x(f) = \frac{a^2 T_b}{4} \text{sinc}^2(f T_b) + \frac{a^2 T_b}{4} \text{sinc}^2(f T_b) \sum_{n=-\infty}^{\infty} \exp(-j 2 \pi n f T_b) \quad \text{--- (4)}$$

Using Poisson's formula

$$\sum_{n=-\infty}^{\infty} \exp(-j 2 \pi n f T_b) = \frac{1}{T_b} \sum_{m=-\infty}^{\infty} \delta\left(f - \frac{m}{T_b}\right) \quad \text{--- (5)}$$

$$\therefore S_x(f) = \frac{a^2 T_b}{4} \text{sinc}^2(f T_b) + \frac{a^2 T_b}{4} \text{sinc}^2(f T_b) \frac{1}{T_b} \sum_{m=-\infty}^{\infty} \delta\left(f - \frac{m}{T_b}\right) \quad \text{--- (6)}$$

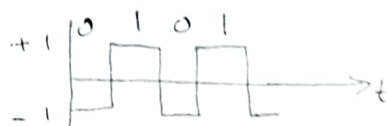


The sinc function $\text{sinc}(f T_b)$ has nulls at $f = \pm 1/T_b, \pm 2/T_b$ as in fig. Therefore the power spectral density is simplified as

$$S_x(f) = \frac{a^2 T_b}{4} \text{sinc}^2(f T_b) + \frac{a^2}{4} \delta(f) \quad \text{--- (7)}$$

The curve a in fig (2) shows the power spectral density of NRZ unipolar format lies b/w dc and the bit rate $(1/T_b)$ of the input data.

② NRZ polar format



Suppose the 0's and 1's of a random binary sequence occur with equal probabilities then, for polar format of the NRZ type we have

$$P(A_k = a) = P(A_k = -a) = 1/2$$

Hence, for $n=0$, autocorrelation function is given as

$$\begin{aligned} E[A_k A_{k-n}] &= E[A_k^2] = (-a)^2 P(A_k = -a) + (a)^2 P(A_k = a) \\ &= a^2 \times \frac{1}{2} + a^2 \times \frac{1}{2} = \frac{2a^2}{2} \\ &= a^2 \end{aligned}$$

Consider next the product $A_k A_{k-n}$ for $n \neq 0$. This has four possible values $[-a \times -a, -a \times a, a \times -a, a \times a]$ i.e. $[a^2, -a^2, -a^2, a^2]$. Assuming that the successive symbols in the binary sequence are statistically independent, these four values occur with a probability of $1/4$. Hence for $n \neq 0$ the autocorrelation is given as

$$E[A_k A_{k-n}] = 2\left(\frac{1}{4}\right) \times a^2 + 2\left(\frac{1}{4}\right) (-a^2) = \frac{a^2}{2} - \frac{a^2}{2} = 0$$

$$\therefore R_A(n) = \begin{cases} a^2 & n=0 \\ 0 & n \neq 0 \end{cases} \quad \text{--- (8)}$$

The Fourier transform of basic pulse $v(t)$ equals

$$V(f) = T_b \text{sinc}(f T_b) \quad \text{--- (9)}$$

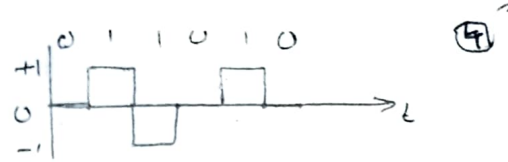
Substituting Eq (9) in Eq (8), yields power spectral density of the NRZ polar format as

$$S_x = \frac{1}{T_b} a^2 T_b^2 \text{sinc}^2(f T_b)$$

$$\therefore \boxed{S_x = a^2 T_b \text{sinc}^2(f T_b)} \quad \text{--- (10)}$$

The curve in fig ② shows power spectral density of NRZ polar format. The power lies inside the main lobe of the sinc-sharped curve, which extends upto the bit rate $1/T_b$.

3) NRZ bipolar format:



In NRZ bipolar format the successive 1's is assigned pulses of alternating polarity. The bipolar format has three levels: a , 0 and $-a$. Then assuming that the probabilities of 0's & 1's occur are equal, then

$$P(A_k = a) = 1/4$$

$$P(A_k = 0) = 1/2, \quad P(A_k = -a) = 1/4$$

The autocorrelation function for $n=0$ is given as

$$\begin{aligned} E[A_k^2] &= (a)^2 P(A_k = a) + (0)^2 P(A_k = 0) + (-a)^2 P(A_k = -a) \\ &= a^2 \times \frac{1}{4} + 0 \times \frac{1}{2} + a^2 \times \frac{1}{4} = \frac{a^2}{2} \end{aligned}$$

for $n=1$, the dibit represented by the sequence (A_{k-1}, A_k) can assume only four possible forms: $(0,0)$, $(0,1)$, $(1,0)$ and $(1,1)$.

The respective values of the product $A_k A_{k-1}$ are $[0, 0, 0, -a^2]$ i.e. $[0 \times 0, 0 \times a, a \times 0, a \times -a]$; the last value results from the fact that successive 1's alternate in polarity. Each of the dibits occurs with probability $1/4$, on assumption that successive symbols in the binary sequence occur in equal probability.

Hence, R_A for $n = \pm 1$ is given as

$$\begin{aligned} E[A_k A_{k-1}] &= 3(0)(1/4) + (-a^2)(1/4) \\ &= -a^2/4 \end{aligned}$$

$$\text{for } n > 1, \quad E[A_k A_{k-n}] = 0$$

$$R_A(n) = \begin{cases} a^2/2 & n=0 \\ -a^2/4 & n=\pm 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{--- (11)}$$

The Fourier transform of basic pulse $v(t)$ is

$$V(f) = T_b \text{sinc}(fT_b) \quad \text{--- (12)}$$

Subs Eq (11) & (12) in Eq (1), with $T = T_b$, the power spectral density of NRZ bipolar format is given by

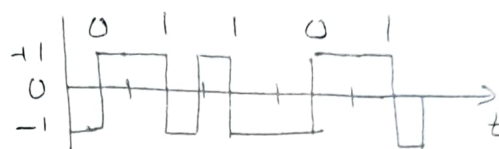
$$\begin{aligned} S_x(f) &= T_b \text{sinc}^2(fT_b) \left[\frac{a^2}{2} - \frac{a^2}{4} [\exp(j2\pi fT_b) + \exp(-j2\pi fT_b)] \right] \\ &= \frac{a^2 T_b}{2} \text{sinc}^2(fT_b) [1 - \cos(2\pi fT_b)] \end{aligned}$$

$$S_x(f) = a^2 T_b \text{sinc}^2\left(f T_b\right) \sin^2\left(\pi f T_b\right) \quad \text{Note:}$$

The curve c in fig ② shows the power spectral density of NRZ bipolar format. The power lies inside a bandwidth equal to the bit rate $1/T_b$, the spectral content of the NRZ bipolar format is relatively small around zero frequency.

② Manchester format:

assume that the binary sequence occur at equal probabilities.



$$\therefore P(A_k = a) = P(A_k = -a) = 1/2$$

Hence for $n=0$, the auto-correlation, R_A is given as

$$E[A_k^2] = (-a)^2 P(A_k = -a) + (a)^2 P(A_k = a) = a^2 \times \frac{1}{2} + a^2 \times \frac{1}{2} = a^2$$

4 for $n \neq 0$, R_A is given as for four values $[a^2, -a^2, -a^2, a^2]$

$$E[A_k A_{k-n}] = 2\left(\frac{1}{4}\right)a^2 + 2\left(\frac{1}{4}\right)(-a^2) = \frac{a^2}{2} - \frac{a^2}{2} = 0$$

$$\therefore R_A(n) = \begin{cases} a^2 & n=0 \\ 0 & n \neq 0 \end{cases} \quad \text{--- (14)}$$

The basic pulse $v(t)$ for the manchester format consists of a doublet pulse of unit amplitude and total duration T_b . The Fourier transform of the pulse equals

$$V(f) = j T_b \text{sinc}\left(\frac{f T_b}{2}\right) \sin\left(\frac{\pi f T_b}{2}\right) \quad \text{--- (15)}$$

Subs Eq (14) & Eq (15) in (1), the power spectral density of the manchester format is given by

$$S_x = a^2 T_b \text{sinc}^2\left(\frac{f T_b}{2}\right) \sin^2\left(\frac{\pi f T_b}{2}\right) \quad \text{--- (16)}$$

curve d in fig ② shows the power spectra of manchester format. The power lies inside a bandwidth equal to $2/T_b$.

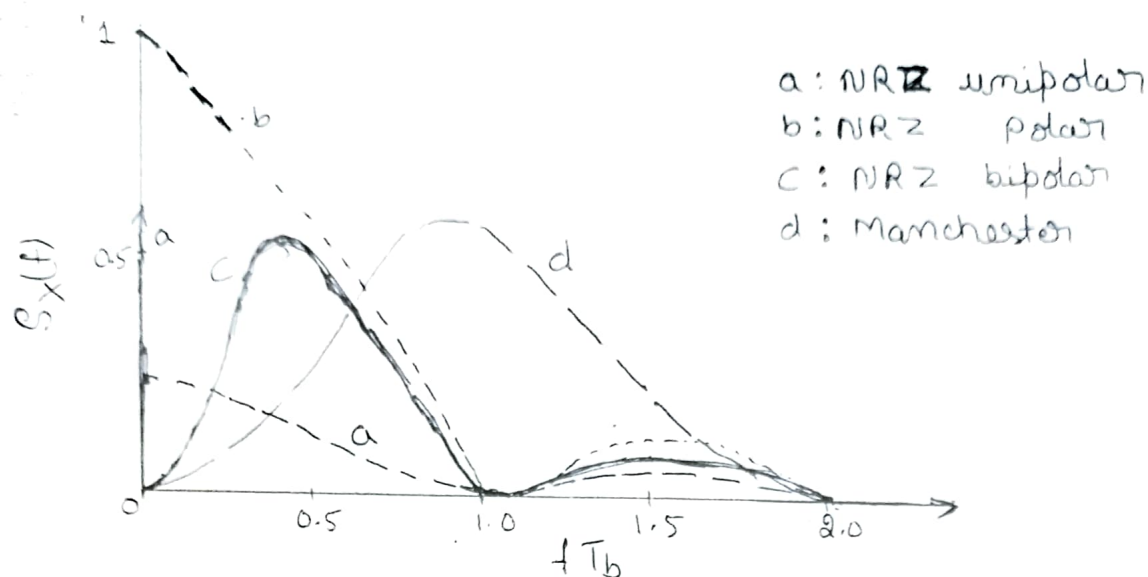


fig 2: Power Spectra of different binary formats.

Intersymbol Interference:

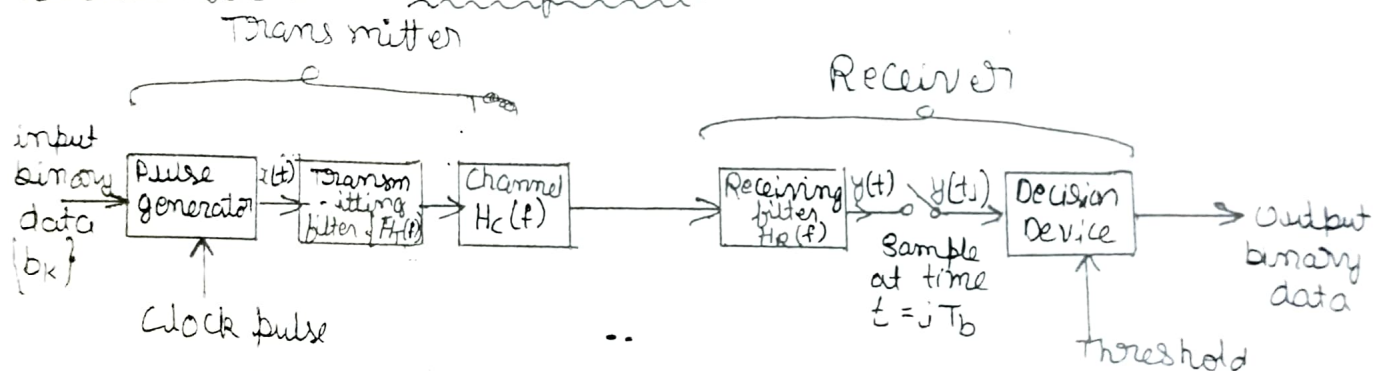


fig: Baseband binary data transmission system

fig shows the basic elements of a baseband binary PAM system. The input signal consists of a binary data sequence $\{b_k\}$ with a bit duration of T_b seconds. This sequence is applied to a pulse generator, producing the discrete PAM signals

$$x(t) = \sum_{k=-\infty}^{\infty} a_k v(t - kT_b) \quad \text{--- (1)}$$

where $v(t)$ denotes basic pulse, normalised such that $v(0) = 1$.

The co-efficient a_k depends on the input data & the type of format used. The PAM signal $x(t)$ passes through a transmitting filter of transfer function $H_T(f)$. The resulting filter o/p defines the transmitted signal, which is passed through a channel of transfer function $H_C(f)$, the channel may be coaxial cable or optical fiber. The channel o/p is passed through a receiving filter of transfer function $H_R(f)$. This filter o/p is sampled synchronously with the transmitter, with the sampling instants being determined by a clock or timing signal that is usually extracted from

the receiving filter output. Finally, the sequence of samples obtained is used to reconstruct the original data sequence by means of a decision device. If the sample exceeds threshold a decision is made in favour of symbol 1, if it does not exceed threshold, a decision is made in favour of symbol 0.

The receiving filter o/p is given as

$$y(t) = \mu \sum_{k=-\infty}^{\infty} a_k p(t - kT_b) \quad \text{--- (2)}$$

where μ is a scaling factor & the pulse $p(t)$ is normalized such that $p(0) = 1$. The pulse $\mu p(t)$ is the response of the cascade connection of the transmitting filter, the channel and the receiving filter, which is produced by the pulse $v(t)$ applied to the input of this cascade connection. Therefore $p(t)$ is related to $v(t)$ in freq domain by writing

$$\mu P(f) = V(f) H_T(f) H_C(f) H_R(f) \quad \text{--- (3)}$$

The normalization of $p(t)$ i.e. $p(0) = 1$ means the total area under the curve of $P(f)$ equals unity.

The receiving filter o/p $y(t)$ is sampled at time $t_i = iT_b$ yielding

$$\begin{aligned} y(t_i) &= \mu \sum_{k=-\infty}^{\infty} a_k p(iT_b - kT_b) \\ &= \mu a_i + \mu \sum_{\substack{k=-\infty \\ k \neq i}}^{\infty} a_k p(iT_b - kT_b) \quad \text{--- (4)} \end{aligned}$$

In Eq (4), the first term μa_i is produced by the i th transmitted bit, where there is no interference because $i = k \Rightarrow p(0) = 1$. The second term represents the residual effect of all other transmitted bits on the decoding of the i th bit; $i \neq k$, this residual effect is called Intersymbol interference (ISI).

ISI arises because of imperfections in the overall frequency response of the system. When a short pulse of T_b seconds duration is transmitted through a band-limited system the frequency components constituting the input pulse are differentially attenuated & differentially delayed by the system. Therefore the pulse appearing at the o/p of the system is dispersed over an interval longer than T_b seconds. Thus when a sequence of short pulses are transmitted through the system, one pulse every T_b seconds the dispersed responses

⑥
 originating from different symbol intervals will interfere with each other, thereby resulting in intersymbol interference. In the absence of ISI, we observe from eq (4) that

$$y(t) = u a_i$$

which shows that, the i th transmitted bit can be decoded correctly. The presence of ISI in the system, introduces errors in the decision device at the receiver o/p.

Nyquist's criterion for distortionless Baseband binary transmission:

The transfer function of the channel and the transmitted pulse shape are specified and the problem is to determine the transfer function of the transmitting and receiving filter's so as to reconstruct the transmitted data sequence $\{b_k\}$. The receiver does this by extracting and then decoding the corresponding sequence of weights, $\{a_k\}$ from the output $y(t)$. Except for a scaling factor, $y(t)$ is determined by the a_k and the received pulse $p(t)$ as in Equation

$$y(t) = u \sum_{k=-\infty}^{\infty} a_k p(t - kT_b) \quad \text{--- (1)}$$

The extraction involves sampling the output $y(t)$ at $t = iT_b$. The decoding requires that the weighted pulse contribution $a_k p(iT_b - kT_b)$ for $k = i$ be free from ISI, due to the overlapping tails of all other weighted pulse where $k \neq i$ results in ISI. This in turn, requires to recover pulse $p(t)$ as shown by

$$p(iT_b - kT_b) = \begin{cases} 1 & i = k \\ 0 & i \neq k \end{cases} \quad \text{--- (2)}$$

By normalisation, $P(0) = 1$. If $P(t)$ satisfies the condition of eq (2), the receiver o/p is $y(t_i) = u a_i$, which is free from ISI. Hence, the condition of eq (2) assures perfect reception in the absence of noise.

From the design point of view, the eq (2) is converted into freq domain. In freq domain we write samples as

$$P_b(f) = R_b \sum_{n=-\infty}^{\infty} P(f - nR_b) \quad \text{--- (3)}$$

where $R_b = 1/T_b$ is the bit rate; $P_g(f)$ is the Fourier transform of an infinite periodic sequence of delta function of period T_b & whose strengths are weighted by respective sample values of $p(t)$. That is $P_g(f)$ is given by

$$P_g(f) = \int_{-\infty}^{\infty} \sum_{m=-\infty}^{\infty} [p(mT_b) \delta(t - mT_b)] \exp(-j2\pi ft) dt \quad \text{--- (3)}$$

Let the integer $m = i - k$. then $i = k$ corresponds to $m = 0$ and $i \neq k$ corresponds to $m \neq 0$. imposing the condition of eq (3) on samples of $p(t)$ in the integral of eq (4)

$$\begin{aligned} \text{we get } P_g(f) &= \int_{-\infty}^{\infty} p(0) \delta(t) \exp(-j2\pi ft) dt \\ &= p(0) \quad \text{--- (5)} \end{aligned}$$

where we have made use of the sifting property of the delta function. Since $p(0) = 1$, by normalization, from eq (3) & (5) that the condition for zero intersymbol interference is satisfied if

$$\sum_{n=-\infty}^{\infty} p(f - nR_b) = T_b \quad \text{--- (6)}$$

Eq (6) constitutes the Nyquist criterion for distortionless baseband transmission in the absence of noise.

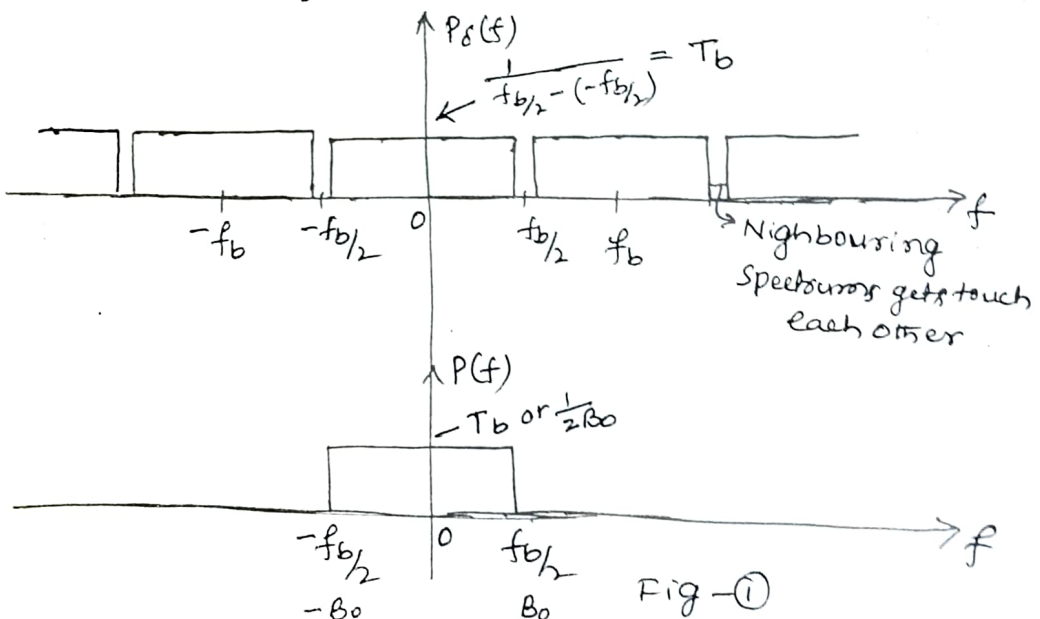
$$\sum_{n=-\infty}^{\infty} P(f - n f_b) = T_b \quad \text{--- (8)}$$

The equation (8) represents frequency domain condition for zero ISI and is called Nyquist pulse shaping criterion for base band transmission.

1. IDEAL SOLUTION TO ISI [IDEAL CHANNEL]

As the expression shown in equation (8) represents the spectrum which get repeated for every f_b is shown in fig-①. Assuming $P(t)$ is a rectangular pulse and its spectrum is represented in 'rect' function

$$\therefore P(f) = \frac{1}{f_b} \text{rect}(f/f_b) \quad \text{--- (9)}$$



According to Fourier transform pairs

$$\text{Sinc}(f_b t) \iff \frac{1}{f_b} \text{rect}(f/f_b)$$

$$\therefore P(t) = \text{Sinc}(f_b t) \quad \text{--- (10)}$$

The bandwidth of pulse can be represented as $B_0 = f_b/2$ 35

$$\therefore f_b = 2B_0, \quad B_0 = \frac{f_b}{2} \quad \text{or} \quad \frac{1}{2T_b} \quad \text{or} \quad \frac{R_b}{2}, \quad R_b = \frac{1}{T_b}$$

$$\therefore p(t) = \text{sinc}(2B_0 t) \quad \text{or} \quad \text{--- (11)}$$

$$p(t) = \frac{\sin(2\pi B_0 t)}{2\pi B_0 t} \quad \text{--- (12)}$$

where $B_0 \Rightarrow$ is the Nyquist bandwidth

As we have considered ideal sinc pulse for zero ISI in base band transmission

According to Fourier transform pair equation (11) is

$$P(f) = \frac{1}{2B_0} \text{rect}(f/2B_0) \quad \text{--- (13)}$$

From fig - (1) $P(f)$ is defined as

$$P(f) = \begin{cases} \frac{1}{2B_0} & \text{for } -B_0 \leq f \leq B_0 \\ 0 & \text{otherwise} \end{cases} \quad \text{--- (14)}$$

Practically ideal sinc pulse can not be generated even though it eliminates ISI completely.

2. PRACTICAL SOLUTION [RAISED COSINE SPECTRUM]

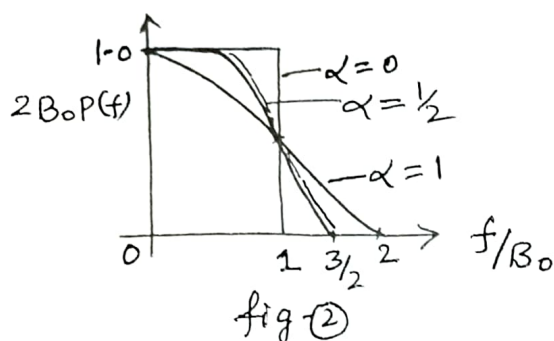
Practically ideal solution is not possible, hence in raised cosine spectrum, the frequency response $P(f)$ decreases to words zero gradually. The raised cosine spectrum is given by

$$P(f) = \begin{cases} \frac{1}{2B_0} & \text{for } -f_1 \leq f \leq f_1 \\ \frac{1}{4B_0} \left\{ 1 + \cos \left[\frac{\pi (|f| - f_1)}{2B_0 - 2f_1} \right] \right\} & \text{for } f_1 < |f| < 2B_0 - f_1 \\ 0 & \text{otherwise} \end{cases} \quad \text{--- (15)}$$

In the above expression the middle term represents BB-13 gradual roll-off from f_1 to $2B_0 - f_1$. The frequency f_1 and Nyquist bandwidth B_0 are related by a roll-off factor ' α '

$$\alpha = 1 - f_1/B_0 \quad \text{--- (16)}$$

The frequency spectrum of $P(f)$ is plotted for different values of ' α ' and is shown in fig (2)



The time domain representation of raised cosine spectrum by taking inverse Fourier transform is

$$P(t) = \text{sinc}(2B_0 t) \frac{\cos(2\pi \alpha B_0 t)}{1 - 16\alpha^2 B_0^2 t^2} \quad \text{--- (17)}$$

From above expression $P(t)$ we observe the following points

- (*) The eqn (17), or pulse function $P(t)$ is a ^{product of} ~~function~~ two functions, which are $\text{sinc}(2B_0 t)$ and decreasing factor $1/t^2$
- (*) Due to $1/t^2$ abrupt transition in the frequency domain are avoided
- (*) The sinc function makes $P(t)$ to zero at the sampling instants

$$t = 0, \pm \frac{1}{2B_0}, \pm \frac{2}{2B_0}, \dots$$

⊗ The time response of raised ~~cos~~ cosine spectrum is shown in fig-③, In this response for all values of 'α' the ISI is eliminated at sampling intervals

⊗ From equation (15), the spectrum has non-zero value upto $2B_0 - f_1$, hence Bandwidth 'B' of RCS

$$B = 2B_0 - f_1$$

$$\text{but } \alpha = 1 - f_1/B_0$$

$$f_1 = B_0 - B_0\alpha = B_0(1 - \alpha)$$

$$\therefore B = 2B_0 - B_0 + \alpha B_0$$

$$\therefore B = B_0 + \alpha B_0$$

$$= (1 + \alpha) B_0$$

Ex: The output of digital computer is at a rate of 64 kbps. if the roll-off factor (i) $\alpha = 1$ (ii) $\alpha = 0.5$ (iii) $\alpha = 0.25$ (iv) $\alpha = 0$ find the bandwidth required to transmit the data in each case

Soln: $f_b = 64 \text{ Kbps}$, $B_0 = \frac{f_b}{2} = 32 \text{ Kbps}$

(i) $B = B_0(1 + \alpha) = 32 \text{ Kbps}(1 + 1) = 64 \text{ kHz}$

(ii) $B = 32 \times 10^3(1 + 0.5) = 48 \text{ kHz}$

(iii) $B = 32 \times 10^3(1 + 0.25) = 40 \text{ kHz}$

(iv) $B = 32 \times 10^3(1 + 0) = 32 \text{ kHz}$

As roll-off factor decreases the bandwidth will also decrease.

CORRELATIVE CODING

BB-14

As we have discussed earlier, the ISI is an undesirable due to degradation in system performance. But adding controlled amount of ISI into the transmitting signal is possible to achieve a bit rate of $2B_0$ bps with a channel bandwidth of $B_0 + \frac{1}{2}$. Such scheme is called correlative coding or partial response signalling scheme. The controlled amount of ISI at transmitter is known and is compensated at the receiver. The correlative coding is implemented by duobinary coding or signalling and modified duobinary signalling.

DUOBINARY SIGNALLING:

The duobinary encoding reduces the maximum frequency of the base band signal. The word 'duo' means doubling of the transmission capacity of the binary system.

Consider a binary sequence $\{b_k\}$ represented as NRZ polar format and is applied to a duobinary encoder, which converts into a three level output as $-2, 0$ and $+2$. The conversion of two level $(1, -1)$ to three level $(-2, 0, 2)$ using duobinary encoder is shown in fig - (1). From diagram the output of duobinary encoder $\{C_k\}$ is given by

$$C_k = b_k + b_{k-1} \quad \text{--- (1)}$$

there the conversion of uncorrelated binary sequence $\{b_k\}$ into an correlated sequence $\{C_k\}$. This is

Uncorrelated $\rightarrow b_k$

Correlated sequence $\rightarrow c_k$

Equivalent to adding controlled amount of ISI

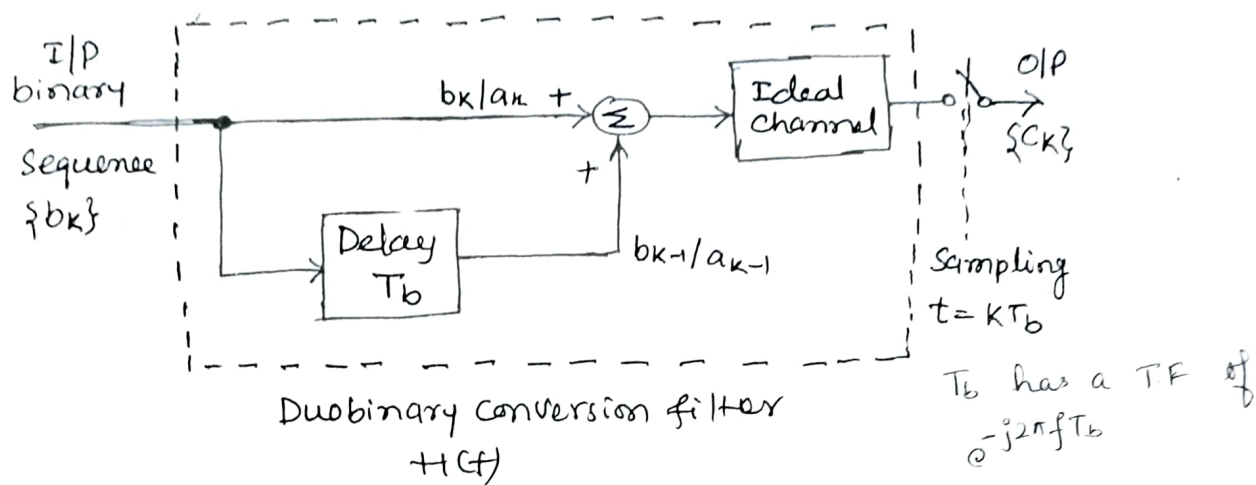


Fig-①

The binary sequence is represented by NRZ polar with $a_k (\pm 1V)$, hence equation ① can also be written as

$$c_k = a_k + a_{k-1} \quad \text{--- ②}$$

$$\text{where } a_k = \begin{cases} +1 & \text{for } b_k = 1 \\ -1 & \text{for } b_k = 0 \end{cases} \quad \text{--- ③}$$

From fig-①, we observe the delay element T_b , which has a transfer function of $e^{-j2\pi f T_b}$. Hence overall transfer function of delay line filter including ideal Nyquist channel is $H(f)$ is given by

$$H(f) = H_c(f) [1 + e^{-j2\pi f T_b}] \quad \text{--- ④}$$

where $H_c(f) \Rightarrow$ Transfer function of channel

$1 + e^{-j2\pi f T_b} \Rightarrow$ Transfer function of delay line filter or dotted box shown in fig-①

$\therefore$ above expression can be rewritten as

$$H(f) = H_c(f) \left[e^{-j\pi f T_b} e^{j\pi f T_b} + e^{-j\pi f T_b} e^{-j\pi f T_b} \right]$$

$$= H_c(f) [e^{j\pi f T_b} + e^{-j\pi f T_b}] e^{-j\pi f T_b}$$

BB-15

$$= H_c(f) 2 \cos(\pi f T_b) e^{-j\pi f T_b} \quad \text{--- (5)}$$

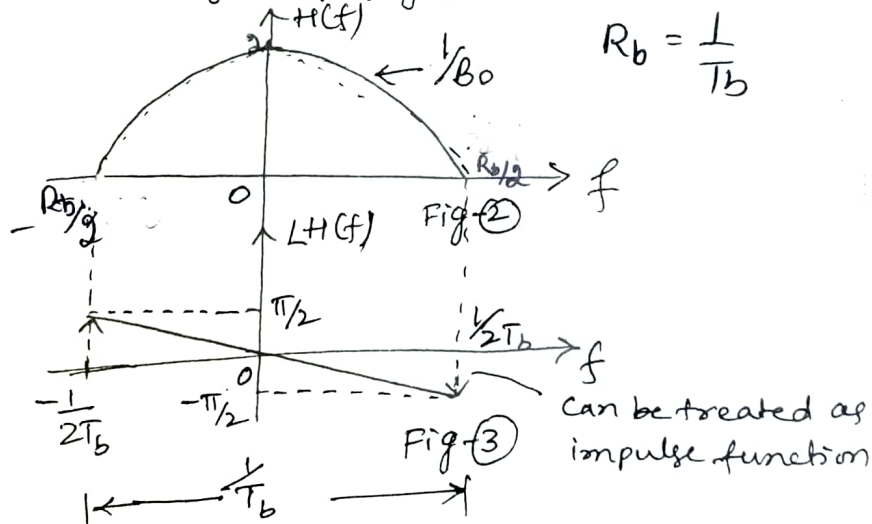
For an ideal channel bandwidth $B_0 = R_b/2$

$$H_c(f) = \begin{cases} 1 & \text{for } f \leq R_b/2 \\ 0 & \text{otherwise} \end{cases} \quad \text{--- (6)}$$

Based on equation (6) equation (5) is defined as

$$H(f) = \begin{cases} 2 \cos(\pi f T_b) e^{-j\pi f T_b} & |f| \leq R_b/2 \\ 0 & \text{otherwise} \end{cases} \quad \text{--- (7)}$$

The amplitude response and phase response of $H(f)$ is shown in fig (2) & fig (3)



At the reconstruction side, let $\hat{a}_k$ represents the estimate of a_k , then we obtain $\hat{a}_k$ as

$$\hat{a}_k = c_k - \hat{a}_{k-1} \quad \text{--- (8)}$$

' c_k ' is the output of transmitter or output of channel. If c_k is received with an error then this

error will propagate in the sequence detected at the receiver output

Ex: The binary data stream 001101001 is applied to the input of a duobinary system. Construct the duobinary encoder output and corresponding receiver output without precoder.

Seqn.	k=0	k=1	k=2	k=3	k=4	k=5	k=6	k=7	k=8
binary seq $\{b_k\}$	0	0	1	1	0	1	0	0	1
polar repr. a_k	-1	-1	+1	+1	-1	+1	-1	-1	+1
$c_k = a_k + a_{k-1}$	-	-2	0	+2	0	0	0	-2	0
Estimated polar o/p $\hat{a}_k = c_k - \hat{a}_{k-1}$	-1	-1	+1	+1	-1	+1	-1	-1	+1
Estimated binary o/p $\hat{b}_k$	0	0	1	1	0	1	0	0	1

b_k

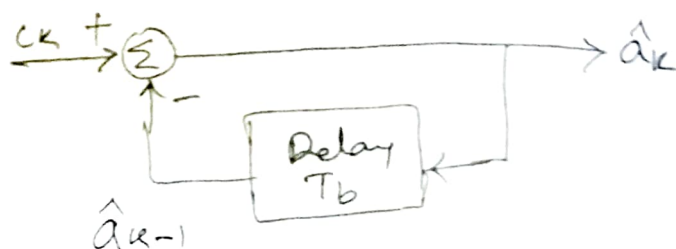
a_k

$$c_k = a_k + a_{k-1}$$

$$\hat{a}_k = c_k - \hat{a}_{k-1}$$

$\hat{b}_k$

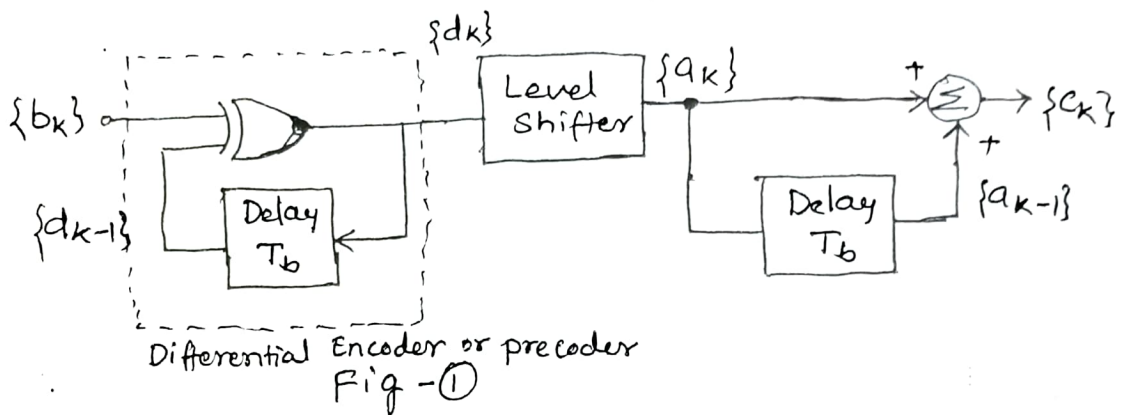
delay



Decoder of Duobinary System

DUOBINARY ENCODER WITH PRECODER [DIFFERENTIAL ENCODER] BB-16

The precoder is used in duobinary encoder to avoid error propagation. The duobinary encoder with precoder is shown in fig-①.



From fig-①, the precoder output $\{d_k\}$ depends on present input bit $\{b_k\}$ and previous output bit $\{d_{k-1}\}$

$$\therefore d_k = b_k \oplus d_{k-1} \quad \text{--- ①}$$

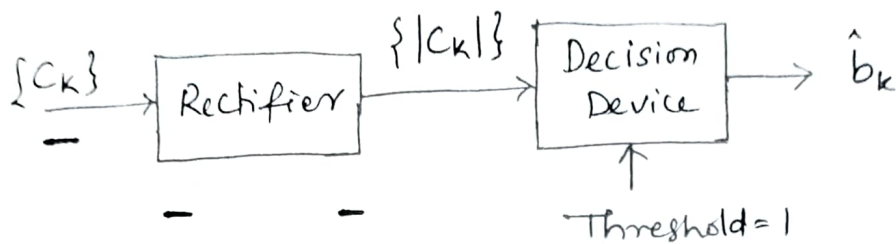
The precoder output $\{d_k\}$ is applied to the level shifting network (NRZ polar representation) and its output is $a_k = \pm 1$ i

$$a_k = \begin{cases} +1 & \text{for } d_k = 1 \\ -1 & \text{for } d_k = 0 \end{cases} \quad \text{--- ②}$$

Therefore the output of encoder is given by

$$c_k = a_k + a_{k-1} \quad \text{--- ③}$$

The duobinary decoder or detector is shown in fig-②



The received three level signal is passing through a rectifier for having the magnitude of C_k and is compared in the decision device for binary sequence b_k .

$$\begin{aligned}
 - \text{ if } |C_k| > 1 & \quad b_k = 0 \\
 |C_k| < 1 & \quad b_k = 1
 \end{aligned}$$

Ex: The binary data stream 0 0 1 0 1 1 0 is applied to the input of duobinary system. Construct duobinary coder output and corresponding receiver output. Assume that there is a precoder at input $d_k = b_k \oplus d_{k-1}$

Soln:- As $\{d_k\}$ depends on $\{b_k\}$ and $\{d_{k-1}\}$, hence to start we have to assume initial bit. The initial output of precoder $\{d_{k-1}\}$ is binary '1' before applying the binary sequence to the precoder

		BB-17							
Sequence		K=-1	K=0	K=1	K=2	K=3	K=4	K=5	K=6
	$\{b_k\}$		0	0	1	0	1	1	0
x	$\{d_{k-1}\}$		1	1	1	0	0	1	0
	$d_k = b_k \oplus d_{k-1}$		1	1	0	0	1	0	0
	$\{a_k\}$	+1	+1	+1	-1	-1	+1	-1	-1
+	$\{a_{k-1}\}$		+1	+1	+1	-1	-1	+1	-1
	$C_k = a_k + a_{k-1}$		+2	+2	0	-2	0	0	-2
	$ C_k $		+2	+2	0	+2	0	0	+2
	$\{\hat{b}_k\}$		0	0	1	0	1	1	0

2. MODIFIED DUOBINARY ENCODING

The modified duobinary technique involves a correlation is over two binary digits. The modified duobinary coder is shown in fig-①. The output of modified duobinary conversion filter is given by

$$C_k = a_k - a_{k-2} \text{ --- ①}$$

If $a_k = \pm 1$, then C_k will have one of three possible levels i.e. 2, 0 and -2 volts

The overall transfer function of the tapped delay line filter connected in cascade with ideal channel shown in fig-① is given by

$$\begin{aligned}
 H(f) &= H_c(f) \left[1 - e^{-j4\pi f T_b} \right] \\
 &= H_c(f) \left[e^{-j2\pi f T_b} e^{j2\pi f T_b} - e^{-j2\pi f T_b} e^{-j2\pi f T_b} \right] \\
 &= H_c(f) e^{-j2\pi f T_b} \left[e^{j2\pi f T_b} - e^{-j2\pi f T_b} \right] \\
 \text{But } \sin \theta &= \left[\frac{e^{j\theta} - e^{-j\theta}}{2j} \right]
 \end{aligned}$$

$$\begin{aligned}
 \therefore H(f) &= H_c(f) e^{-j2\pi f T_b} [2j \sin 2\pi f T_b] \\
 &= 2j H_c(f) \sin 2\pi f T_b e^{-j2\pi f T_b} \quad \text{--- (2)}
 \end{aligned}$$

Therefore overall frequency response of equation (2) and its phase response is shown in fig (2)

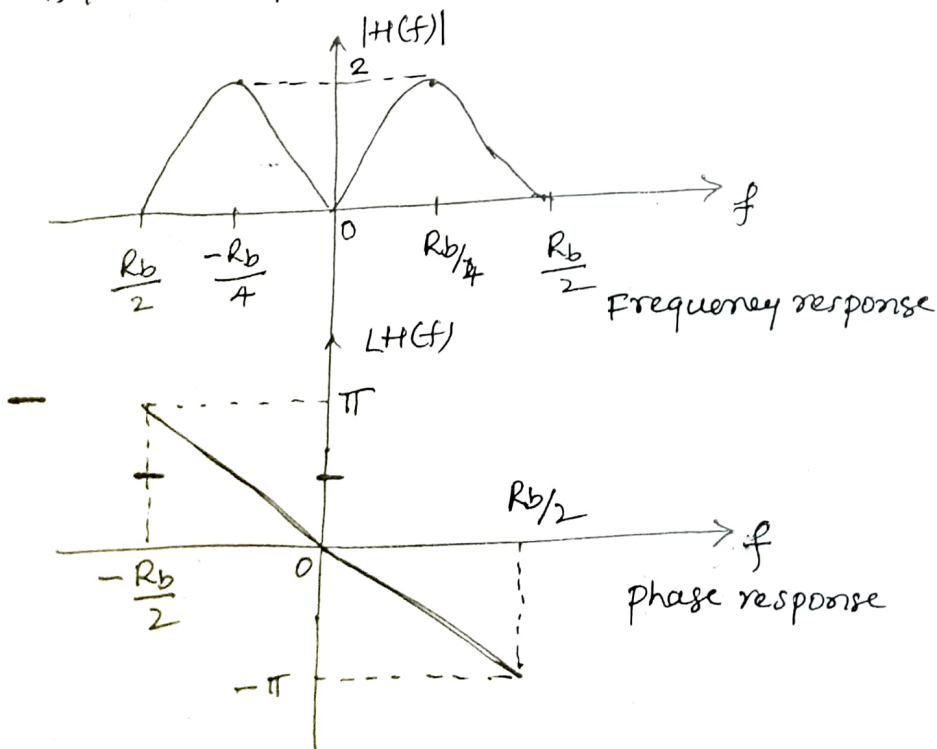


fig (2)

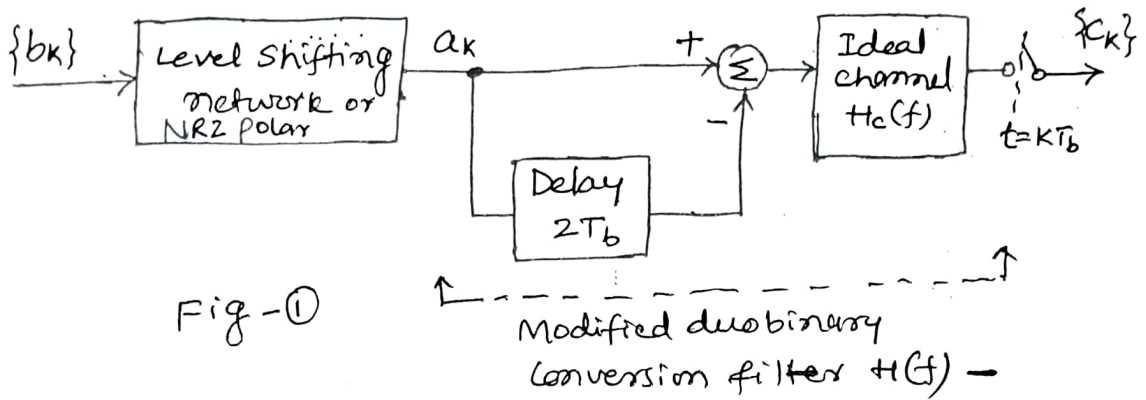
$$H(f) = \begin{cases} 2j \sin 2\pi f T_b e^{-j2\pi f T_b} \\ 0 \end{cases}$$

$$|f| \leq R_b/2$$

otherwise

BB-18

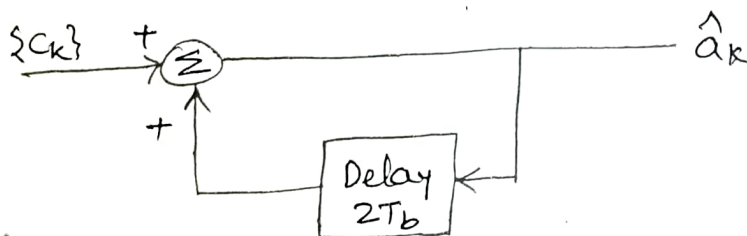
(3)



At the decoder, let $\hat{a}_k$ is the estimate of a_k , then we obtain $\hat{a}_k$ as

$$\hat{a}_k = c_k + \hat{a}_{k-2} \quad \text{--- (4)}$$

The decoder of modified duobinary system is shown in fig-(3)



The $\hat{a}_k$ is sampled for every bit $t = kT_b$ and is compared with decision device

$$\hat{a}_k > T_h \quad b_k = 1$$

$$\hat{a}_k < T_h \quad b_k = 0$$

Ex: The binary data 011100101 are applied to the input of modified duobinary system.

(i) Construct the modified duobinary coder output and corresponding receiver output without pre-coder

(ii) Suppose that due to error in transmission, the level produced by the third digit is reduced to zero. Construct a new receiver output

OP
(1)

Seq K=-2 K=-1 K=0 K=1 K=2 K=3 K=4 K=5

Seq	K=-2	K=-1	K=0	K=1	K=2	K=3	K=4	K=5	K=6	K=7	K=8
$\{b_k\}$			0	1	1	1	0	0	1	0	1
$\{a_k\}$	+1	+1	-1	+1	+1	+1	-1	-1	+1	-1	+1
$\{a_{k-2}\}$			+1	+1	-1	+1	+1	+1	-1	-1	+1
$\{c_k\} = a_k - a_{k-2}$			-2	0	+2	0	-2	-2	+2	0	0
$\hat{a}_{k-2}$			+1	+1	-1	+1	+1	+1	-1	-1	+1
$\hat{a}_k = c_k + \hat{a}_{k-2}$	+1	+1	-1	+1	+1	+1	-1	-1	+1	-1	+1
$\hat{b}_k$			0	1	1	1	0	0	1	0	1

BB-19

(ii) Due to error in transmission, the level produced by 3rd digit is reduced to zero

Seq	-2	-1	0	1	2	3	4	5	6	7	8
c_k without error			-2	0	+2	0	-2	-2	+2	0	0
c_k with error			-2	0	0	0	-2	-2	+2	0	0
$\hat{a}_{k-2}$			+1	+1	-1	+1	+1	+1	-3	-1	-1
$\hat{a}_k =$ $c_k + \hat{a}_{k-2}$	+1	+1	-1	+1	-1	+1	-3	-1	-1	-1	-1
$\hat{b}_k$			0	1	0	1	0	0	0	0	0

The reconstructed sequence shown in the last row contains error up to last digit. This shows that error error propagates in this system. Hence a precoder must be used at the input of duobinary encoder to avoid error propagation.

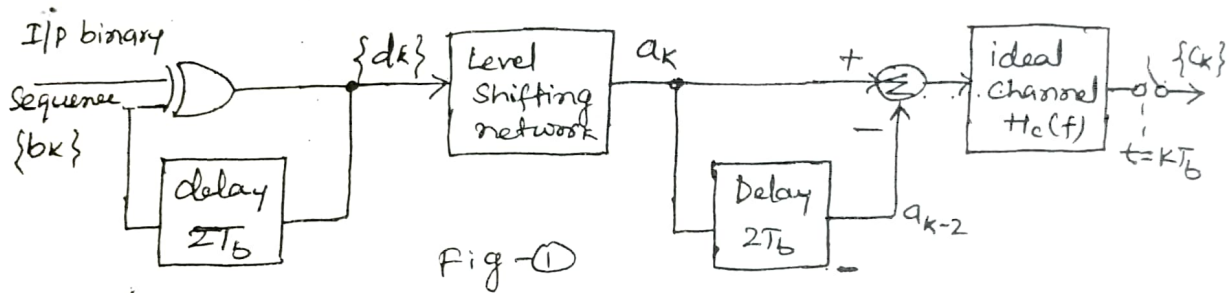
M

..
..

MODIFIED DUOBINARY ENCODING WITH PRECODER

The block diagram of modified duobinary encoder with precoder is shown in fig-①. As precoder performs modulo-2 addition [EX-OR operation] on binary bits separated by $2T_b$. The output of precoder is given by

$$d_k = b_k \oplus d_{k-2} \quad \text{--- ①}$$



The sequence $\{d_k\}$ applied to level shifting network [NRZ polar] and output of it is represented as a_k

$$\therefore a_k = \begin{cases} -1 & \text{for } d_k = 0 \\ +1 & \text{for } d_k = 1 \end{cases} \quad \text{--- ②}$$

Finally the output of modified duobinary encoder with precoder $\{c_k\}$ is given by

$$\{c_k\} = a_k - a_{k-2} \quad \text{--- ③}$$

$\{c_k\}$ generated three levels such as +2, -2 and '0'

At the reconstruction side, the modified duobinary decoder block diagram is shown in fig-②

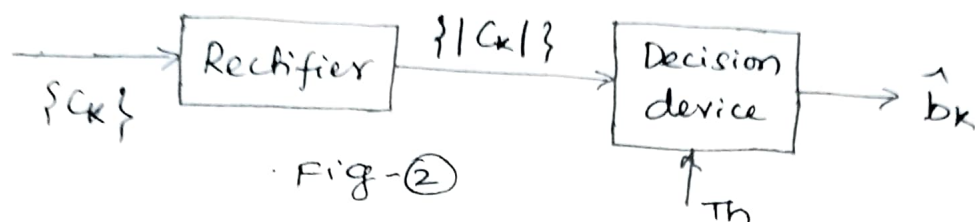


Fig-(2)

The decoder produces output binary digits based on comparison of $|C_k|$ and threshold value of decision device

$$\text{If } |C_k| > 1 \quad \hat{b}_k = 1$$

$$|C_k| < 1 \quad \hat{b}_k = 0$$

Next

GENERALIZED FORM OF CORRELATIVE CODING

The duobinary and modified duobinary systems have correlation span of 1 binary digit $[T_b]$ and 2 binary digits $[2T_b]$ respectively. This can be generalized using tapped delay line filters with tap weights of $w_0, w_1, w_2, w_3, \dots, w_{N-1}$. This scheme is shown in fig-(1).

Specifically a correlative sample C_k is obtained from a superposition of 'N' successive input sample values b_k is given by

$$C_k = \sum_{n=0}^{N-1} w_n b_{k-n}$$

where, $n=1$

$$w_0 = +1$$

$$w_1 = +1$$

} duobinary scheme

for $n=2$

$$w_0 = +1$$

$$w_1 = 0$$

$$w_2 = -1$$

} modified duobinary system.

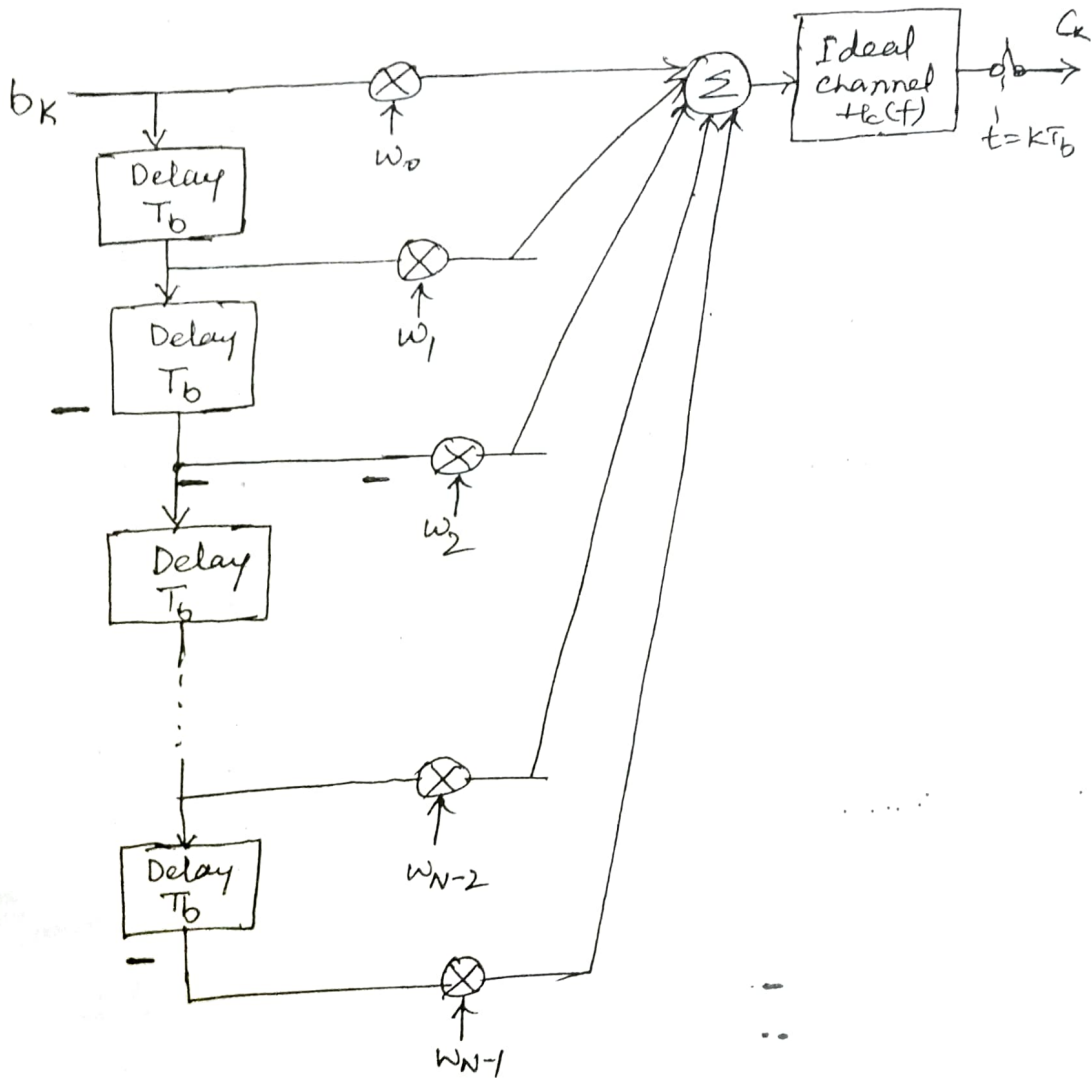


Fig-①: Tapped delay line filter